Investigating the use of Wasserstein distance for Regularizing Vector Auto-Regressive (VAR) Models

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Introduction

Vector Autoregressive (VAR) models provide a framework for forecasting the relationships of multiple time series variables simultaneously (Lütkepohl, 2005). They expand upon the univariate auto-regressive (AR) models by allowing for multivariate time series. Specifically, a VAR(p) model allows for each variable to be regressed on p of its own lags as well as p lags of all other variables. This allows for VAR models to capture complex relationships where variables mutually influence each other over time. Estimation of VAR(p) models often involves ordinary least squares (OLS) on all p equations, sometimes with penalization to enforce sparsity and avoid overfitting (Basu & Michailidis, 2015). Overfitting is an especially prevalent issue in time series data, due to the low number of time points relative to the amount of variables needed to estimate. In both psychology and neuroscience, VAR models have become increasingly popular for characterizing temporal data at both individual and group levels. For example, network models based on VAR have been used in modeling and quantifying the dynamics between affective states (Bringmann et al., 2016). In neuroimaging, multi-task regression models that build upon VAR have been used to perform source localization in magnetoencephalography, applying some similarities between people (Janati et al., 2019).

However, VAR models in their current state do not seem to provide the boost in modeling capabilities that some may intuitively feel they should: studies have found that VAR(1) models suffer in performance with small samples (Bulteel et al., 2018). In this paper, we hope to use a metric from optimal transport called Wasserstein distance to regularize the VAR and improve forecasting performance.

VAR

The VAR(p) model (for one person) can be expressed as:

$$Y_t = c + \Phi_1 Y_{t-1} + \Phi_2 Y_{t-2} + \cdots + \Phi_p Y_{t-p} + \epsilon_t. \tag{1}$$

where Y_t is a $d \times 1$ vector containing all d variables at time point t (what we try to forecast), $\Phi_1, \ldots, \Phi_p$ are $d \times d$ coefficient matrices representing the influences of all variables on Y_t from time $t-1$ to $t-p$, and $Y_{t-1}, \ldots, Y_{t-p}$ representing the past variables from time $t-1$ to $t-p$. c and ϵ_t are the overall intercept and the errors for time t, respectively.

The goal is to solve for the coefficient matrices $\Phi_1, \ldots, \Phi_p$ given the observed set of data points $Y_1, \ldots, Y_t$. This is commonly done using OLS. Without loss of generality, let us assume that Y_t has mean 0. The OLS optimization problem (for one person) is as follows:

$$\Phi_1, \ldots, \Phi_p \ = \underset{\Phi_1, \ldots, \Phi_p}{\arg\min} \sum_{t=p+1}^{T} \left\| Y_t - \Phi_1 Y_{t-1} - \Phi_2 Y_{t-2} - \cdots - \Phi_p Y_{t-p} \right\|_2^2. \tag{2}$$

However, this estimation of the coefficient matrix can suffer in cases of high number of parameters and generally low amounts of data. A promising approach outlined in (Fisher et al., 2022) called multiVAR, accounts for the general similarities between people by applying a penalty that enforces some closeness to the group mean. In this paper, we introduce an approach to estimate coefficient matrices that uses a technique from optimal transport (OT). OT aims to find methods for quantifying the ‘distance’ between two probability distributions. We apply a method from (Janati et al., 2019), using the Wasserstein distance between distributions for regularizing our multi-task regression task.

The Wasserstein distance is calculated using a *ground metric* matrix M, which encodes individual distances between variables. Using M, we can find the cost to move from one distribution to another, denoted in the *transport plan* P (**??**). In (Janati et al., 2019), distance was calculated using the physical distance between areas on the brain. In our case, we aim to use prior information about the psychological variables analyzed. For example, we know that ‘happy’ and ‘sad’ are farther apart than ‘sad’ and ‘depressed’. A more objective and numeric measure of the distances of these words can be found using word embeddings, which are created and utilized by various Large Language Models (LLM's) (Kjell et al., 2023). On a high level, the goal of the Wasserstein distance is to promote similarity between the coefficient matrices of various people, by minimizing the Wasserstein distance between their individual parameters.

The objective function that we aim to minimize is as follows, given K tasks and T time points. We use $p = 1$, only considering the variables from the previous time point to predict the next point:

$$\{\Phi^{(k)}\}_{k=1}^{K} = \underset{\{\Phi^{(k)}\}_{k=1}^{K}}{\arg\min} \left[\sum_{k=1}^{K} \sum_{t=2}^{T} \left\| Y_t^{(k)} - \Phi^{(k)} Y_{t-1}^{(k)} \right\|_2^2 \; + \; J\Big(\Phi^{(1)}, \ldots, \Phi^{(K)}\Big) \right]. \tag{3}$$

Where J represents the regularization penalty, encoding the Wasserstein distance which we apply column-wise on Φ for its d columns:

$$J\Big(\Phi^{(1)}, \ldots, \Phi^{(K)}\Big) = \min_{\bar{\Phi}} \frac{\alpha}{K} \sum_{k=1}^{K} \sum_{j=1}^{d} \Delta\Big(\Phi_{\cdot j}^{(k)}, \; \bar{\Phi}_{\cdot j}\Big), \tag{4}$$

$\bar{\Phi}$ is the barycenter, or the 'average' that captures the typical participant. A penalty is enforced encouraging the Wasserstein distance between any participant's coefficient matrix and the 'average' participant to be minimized, with α denoting the strength of the penalty. Δ represents the Wasserstein distance between the two vectors. As this distance is only defined on non-negative vectors, we break up each vector into its positive and negative parts, such that $x_+ = \max(x, 0)$.

$$\Delta(x, y) \coloneqq W(x_+, y_+) \; + \; W(x_-, y_-). \tag{5}$$

Finally, we define the Wasserstein distance function W:

$$W(x, y) = \min_{P \in \mathbb{R}_+^{d \times d}} \Big[\langle P, M \rangle - \varepsilon \, E(P) + \gamma \, \mathrm{KL}(P\mathbf{1} \,\|\, x) + \gamma \, \mathrm{KL}(P^\top \mathbf{1} \,\|\, y) \Big]. \tag{6}$$

Where M is the ground metric, P is the $d \times d$ transport plan, γ the level of penalization for unbalanced transports, and ϵ the level of penalty for unsmooth plans.

Here $\langle P, M \rangle$ denotes the Frobenius inner product,

$$\langle P, M \rangle = \sum_{i,j} P_{ij} M_{ij}.$$

The entropy term is the Shannon entropy,

$$E(P) = -\sum_{i,j} P_{ij} \log P_{ij} \; + \; \sum_{i,j} P_{ij}.$$

$\mathrm{KL}(x \,\|\, y)$ denotes the discrete Kullback–Leibler divergence:

$$\mathrm{KL}(x \,\|\, y) = \sum_i x_i \log \frac{x_i}{y_i} + \sum_i (y_i - x_i).$$

These definitions specify the penalty terms used in the unbalanced optimal transport formulation.

Methods

Simulations

Before applying our Wasserstein regularized multi-task VAR approach to real data, we conducted a simulation study to evaluate whether the method behaves sensibly in controlled settings. We generated synthetic data from a VAR(1) system with $d = 5$ variables and specified a pre-defined ground metric M, which encodes pairwise distances between variables. This ground metric was then used to control the structure of the VAR coefficient matrix $\Sigma_\Phi = \exp(\beta * M)$ (applied elementwise to M). For each task k, the ground truth coefficient matrix $\Phi^{(k)}$ was sampled from a multivariate normal distribution with covariance Σ_Φ, and a noise covariance matrix $\Psi^{(k)}$ was also generated. Data was simulated by iteratively sampling $Y_t^{(k)} = \Phi^{(k)} Y_{t-1}^{(k)} + \epsilon_t^{(k)}$, where ϵ_t follows a multivarate normal distribution with mean zero and covariance $\Psi^{(k)}$.

Our goal was to test whether as the number of time points grows large, the Wasserstein regularized estimator stabilizes toward zero loss (in the same way that OLS does). This property allows us to verify that the method we chose is valid with large samples.

We simulated data with $T = 1000$ time points and fit the Wasserstein regularized VAR under various α values, running a cross-validation to choose the best α. Across conditions, the training loss decreased rapidly and stabilized near zero, mirroring the behavior of the OLS estimator. This shows that the Wasserstein penalty can be used to recover the true coefficient matrices in case of large time points. With this validation in place, we proceeded to evaluate the method on empirical EMA data, as seen below.

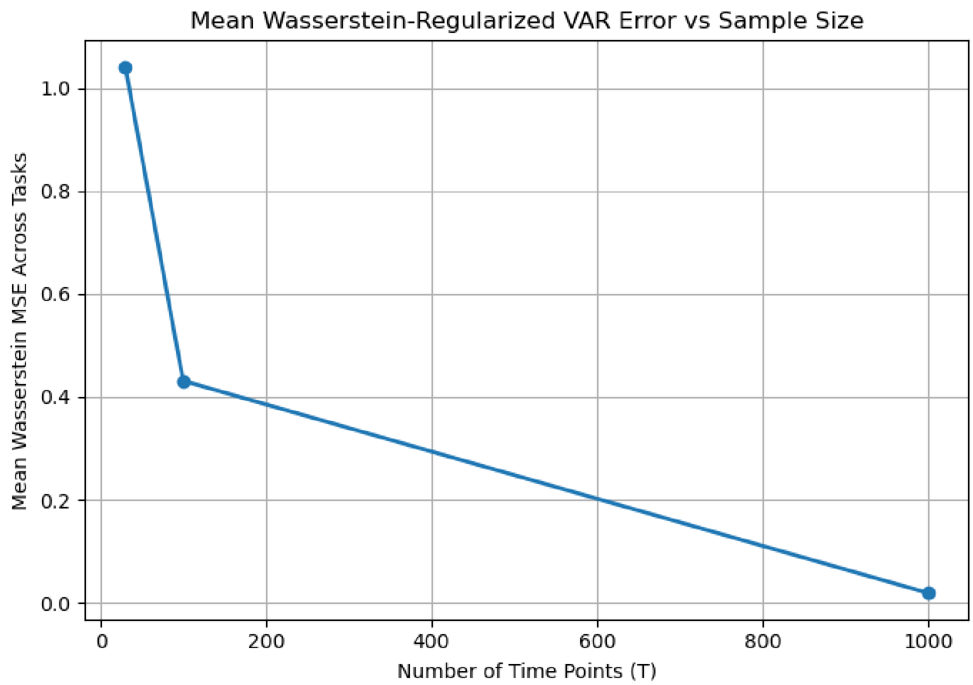

Figure 1

Mean errors in Wasserstein distance on simulated data with respect to amount of time points.

Empirical Application

To evaluate the performance of our Wasserstein regularized multi-task VAR approach, we conducted an empirical study on real world data. We used ecological momentary assessment (EMA) data (Fried et al., 2021) which followed 80 adults during the early Covid period, from March 16th to March 29th, 2020 (two weeks, or fourteen days in total). Participants were prompted to fill out a survey four times a day (56 total survey responses), with each question assessing a range of emotional states, reported on an ordinal scale. This data models time series data, in which we can use the data from previous points to predict the data at the next point.

Preprocessing

We took the data for 8 out of a total 14 questions asked on the survey. This was done to make our experiment a proof of concept as well as to keep the dimensions of the VAR manageable. We chose 8 questions that we hypothesized had some semantic relationship with each other, and these served as our d variables for the study. They are as follows:

Q1: I found it difficult to relax

Q2: I felt very irritable

Q3: I was worried about different things

Q4: I felt nervous, anxious, or on edge

Q11: I spent __ minutes on meaningful, offline social interaction

Q12: I spent __ minutes using social media to pass the time

Q15: I spent __ minutes outside

Q18: I spent __ minutes at home

All questions had five answer options, with Q1, Q2, Q3, and Q4 on a Likert scale, with responses ranging from ‘Not at all’ to ‘Extremely’. Q11, Q12, Q15, and Q18 also were measured on an ordinal scale, with the available categories listed as ‘0 minutes’, ‘1-15 minutes’, ‘15-60 minutes’, ‘1-2 hours’, ‘>2 hours’. As is common with real data, there were several missing survey responses, which could be due to a lack of a response or a response that arrived too late. For our final dataset, we took all participants that had at least 87.5% of the surveys completed, and used linear interpolation to impute the remaining missing data (Xu & Xu, 2022). After these steps, we were left with a total of 42 unique participants in our study, each with 56 total data points.

Ground metric construction

A key innovation of our approach is the use of psychologically meaningful distances between emotion variables as a ground metric for Wasserstein regularization. We obtained ground metric matrix M using word embedding models, which characterize the distance between all pairs of these 8 variables, (size $d \times d$) (Devlin et al., 2019; Liu et al., 2019; Yang et al., 2019).

Importantly, we hypothesized that the columns of matrix Φ would be similar among tasks. That is, we hypothesize that the effect of the same predictors on different outcomes would be similar among people (i.e. assuming that the relationship between ('happy' $\rightarrow$ 'sad') and ('happy' $\rightarrow$ 'happy') is captured by the ground metric). We iterate with blockwise descent when applying the ground metric columnwise, updating the coefficient matrix values until convergence (difference of ≤ 0.001 between consecutive Φ matrices) or a set amount of passes through all columns of the matrix (which we tentatively set to be 25).

The differences in the values of the three matrices are located in Figure 1 below. To test the differences between various models, we compare the forecasting results between three different M's, modeled by BERT, RoBERTa, and XL-Net.

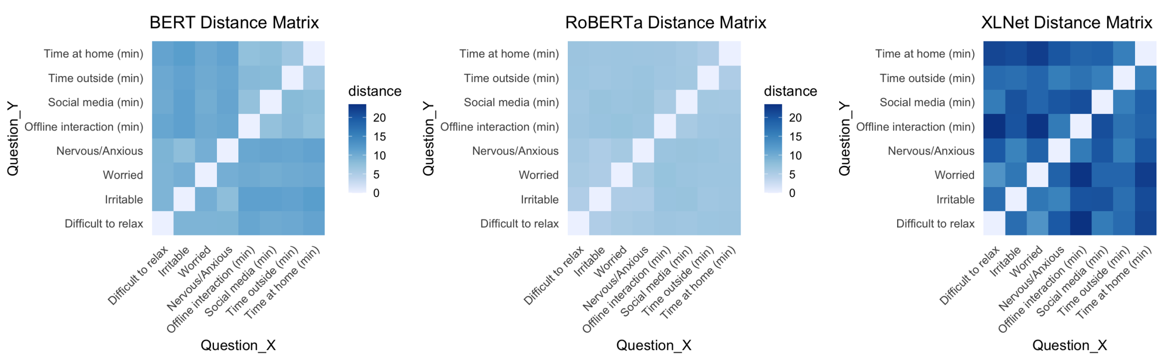

Figure 2

Heatmaps of the three word embedding based ground metric matrices.

Models/Approach

To compare the effect of our approach, we compare it with the standard OLS approach, where the Wasserstein regularization penalty $\alpha = 0$.

To choose the best α, we used a blocked cross-validation approach. We used 5 folds, holding out 20% of the time series at each fold, with the held out segment remaining consecutive in time. Forecasting was done one time point into the testing set, which served as our metric for choosing the best alpha. Errors were summed across all 8 variables predicted for all people, and then across all folds, to choose the best alpha for the full dataset.

After the best alpha was chosen (which could vary for any of the three ground metric matrices M), we separated the full dataset into training and testing, using the final 5 data points as testing. After forecasting this testing set (using the standard VAR(1) prediction $\hat{Y}_{t+1} = \Phi Y_t$), we compared the results between predicting using the Φ's generated by the multi-task Wasserstein fit versus the standard OLS fit with $\alpha = 0$.

Results

We evaluated whether adding a Wasserstein regularization term improves various step forecasts compared to the standard VAR(1) model estimated with OLS ($\alpha = 0$). For each participant and horizon $h = 1, \ldots, 5$, we computed the difference in mean squared forecast error between the best cross validated Wasserstein model and the OLS baseline ($\mathrm{MSE}_{\mathrm{best}} - \mathrm{MSE}_{\alpha=0}$). The cross-validation procedure for selecting the best alpha consistently selected alpha values that were strictly positive (even though $\alpha = 0.0$ was in the pool), indicating that Wasserstein regularization may provide some boost in predictive power.

Negative values for the error difference depict that the Wasserstein regularized model achieved lower forecast error, which may support our hypothesis. The resulting distributions are summarized in Figures 3-7 with boxplots for each horizon and ground metric, where each boxplot is comprised of all 42 participants used in our study.

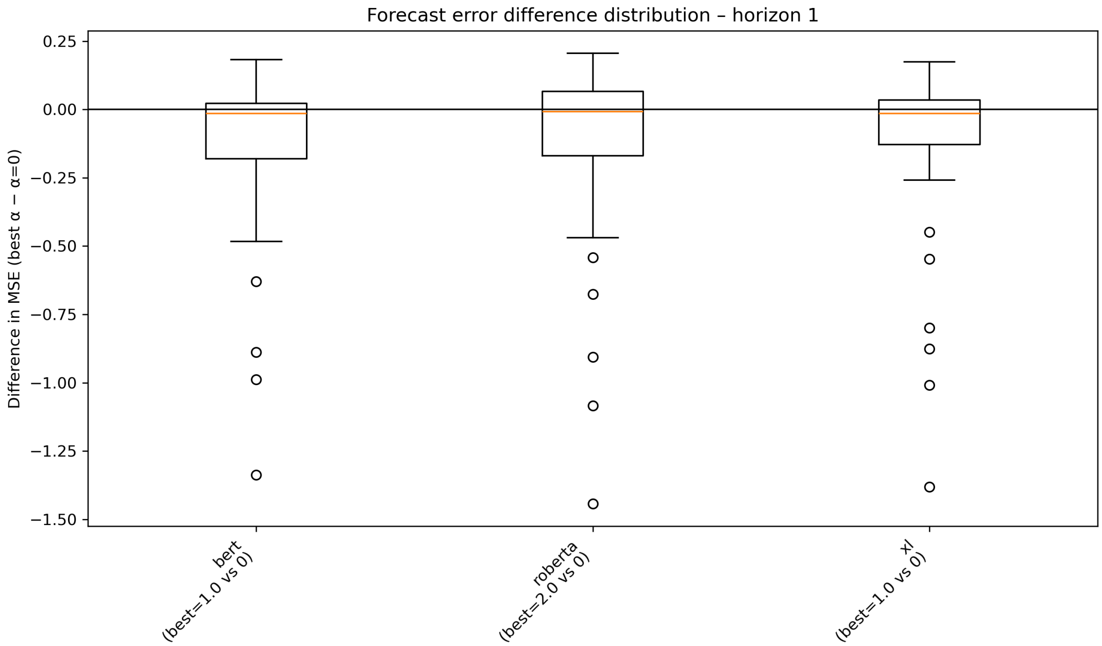

Figure 3

Forecast error difference distribution at horizon $h = 1$. Each boxplot shows the distribution of $MSE_{best} - MSE_{\alpha=0}$ across participants for specified ground metric (BERT, RoBERTa, XLNet). Negative values indicate an improvement over the unregularized VAR(1) baseline.

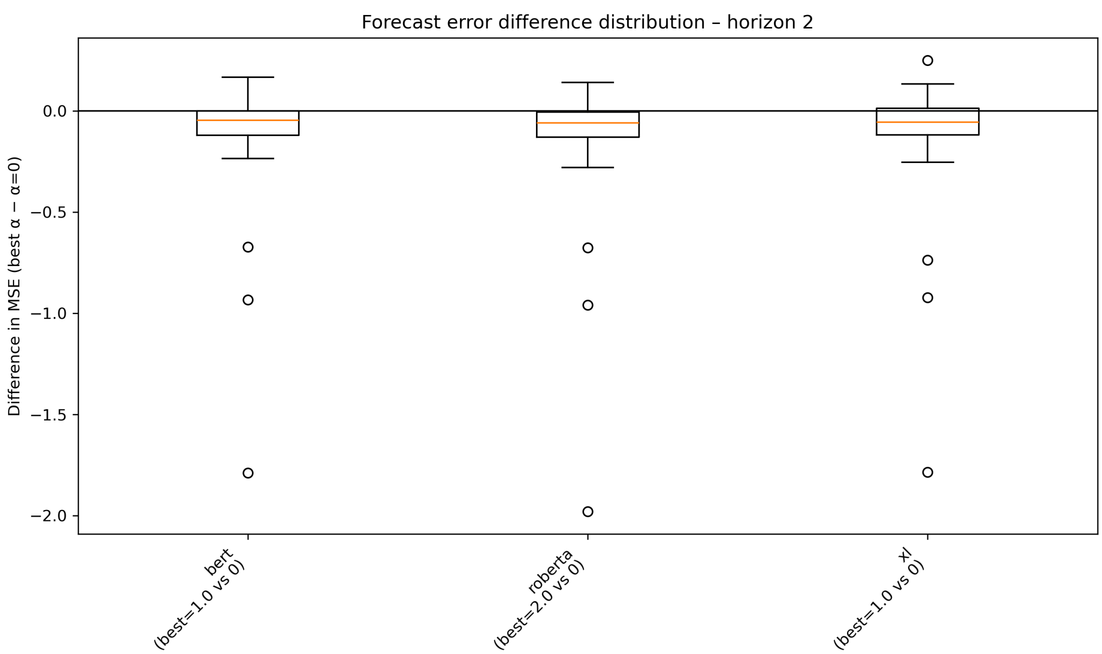

Figure 4

Forecast error difference distribution at horizon $h = 2$. Conventions as in Figure 3.

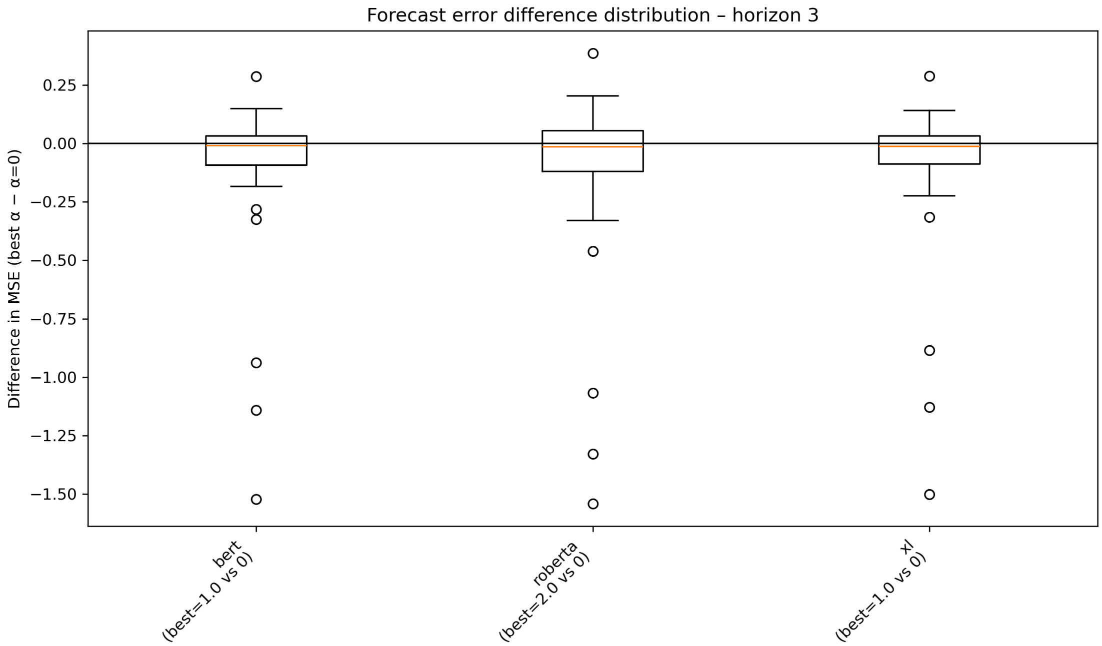

Figure 5

Forecast error difference distribution at horizon $h = 3$. Conventions as in Figure 3.

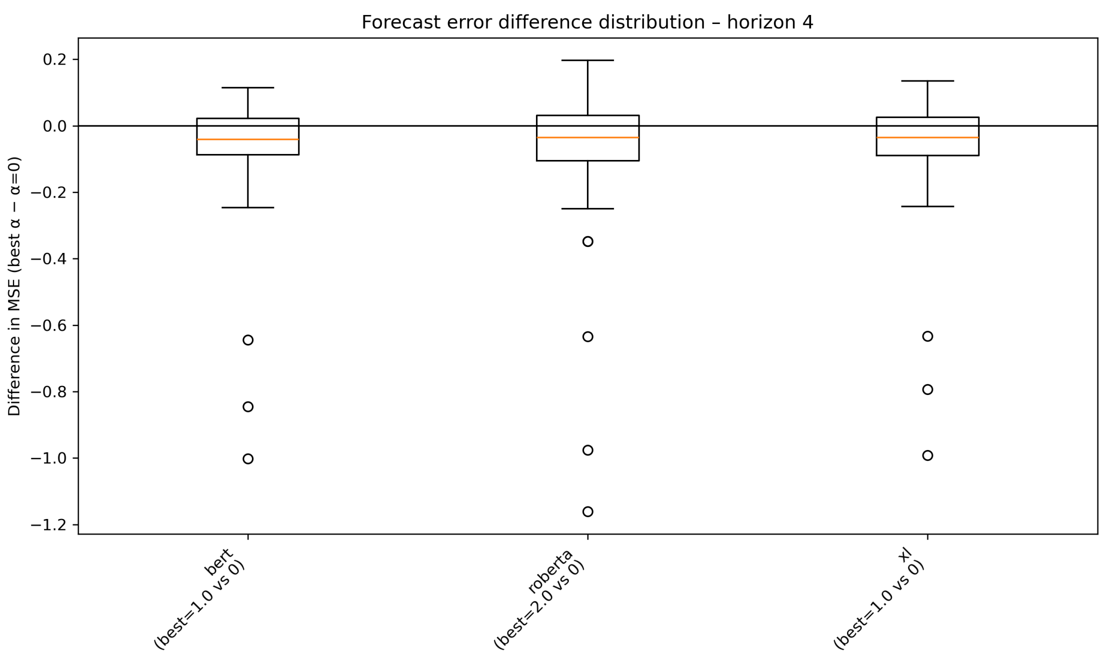

Figure 6

Forecast error difference distribution at horizon $h = 4$. Conventions as in Figure 3.

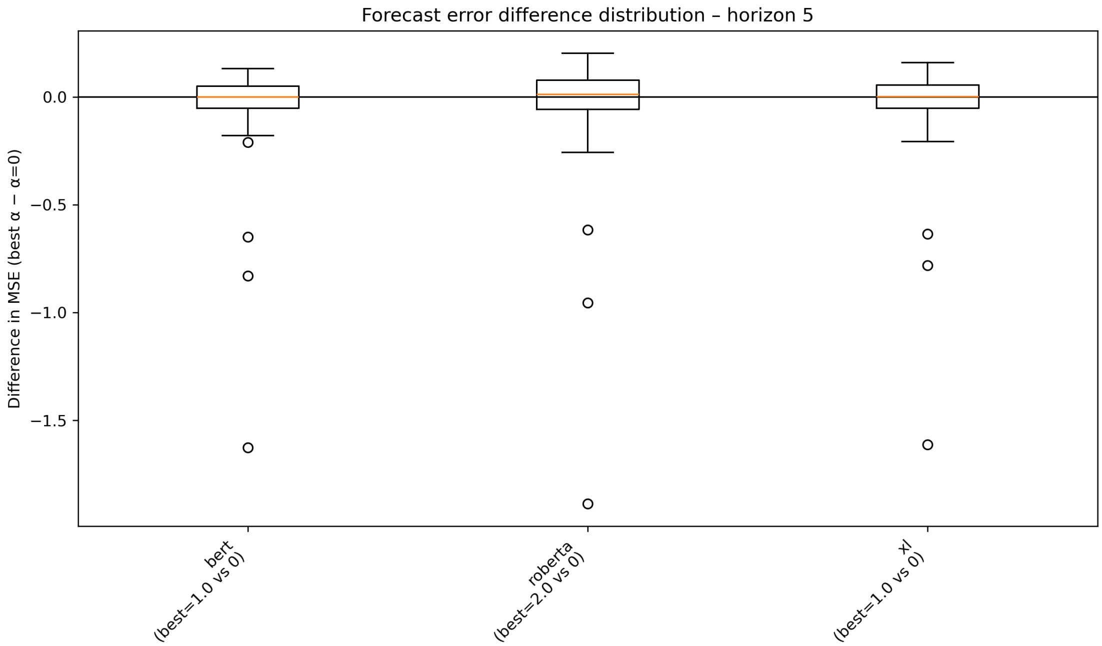

Figure 7

Forecast error difference distribution at horizon $h = 5$. Conventions as in Figure 3.

Forecast error distributions across horizons

Across horizons $h = 1, \ldots, 4$, the medians of the error differences for all three embedding based ground metrics are slightly negative. This suggests that on average, incorporating Wasserstein regularization yields modest yet consistent improvements in forecast accuracy over the standard VAR(1) approach. For most horizon and ground metric combinations, the bulk of the distribution is concentrated in a band around zero, which indicates that the effect size is small for the majority of participants. However, there are also several outliers present in each distribution, which shows that for a small subset of participants that the Wasserstein regularized model substantially outperforms the baseline.

For horizon $h = 5$, the results show a median centered more around zero, with the few outliers still remaining. This finding is typical within time series forecasting, as errors compound over previous horizons, and the model is working with older data that may not be as important after five points have been assessed, especially in modeling a VAR(1) model.

Per-task error patterns

To examine how performance varied at the individual level, we plotted per-task MSE differences between the best Wasserstein regularized model and the standard OLS baseline for each ground metric and horizon (Figures 8-12). These bar plots reveal patterns that are less visible in the aggregated boxplots.

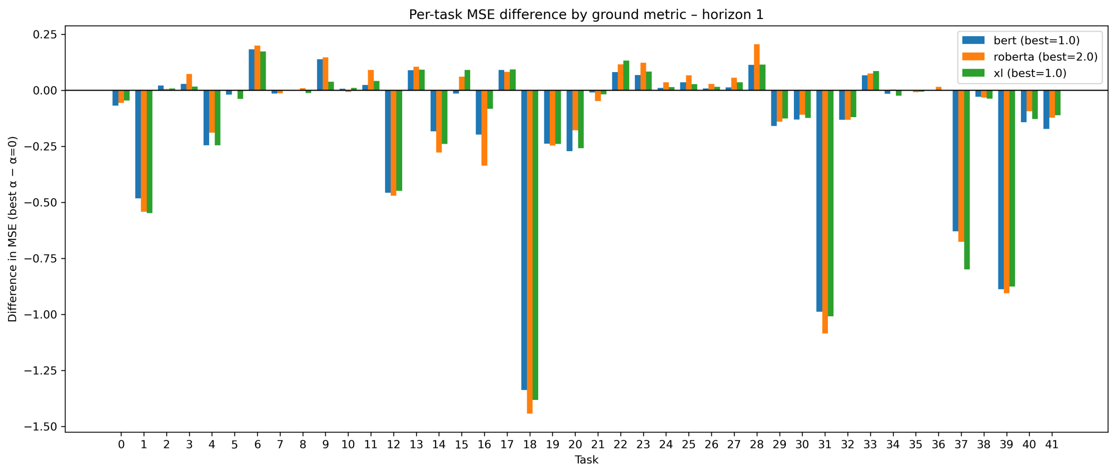

Figure 8

Per-task MSE difference by ground metric for horizon $h = 1$. Negative values indicate improved performance under the Wasserstein regularized model.

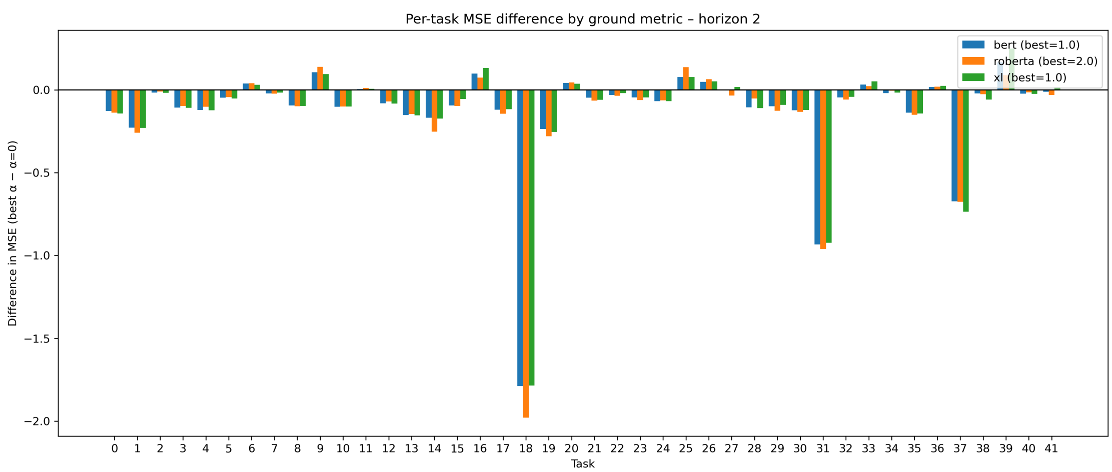

Figure 9

Per-task MSE difference by ground metric for horizon $h = 2$.

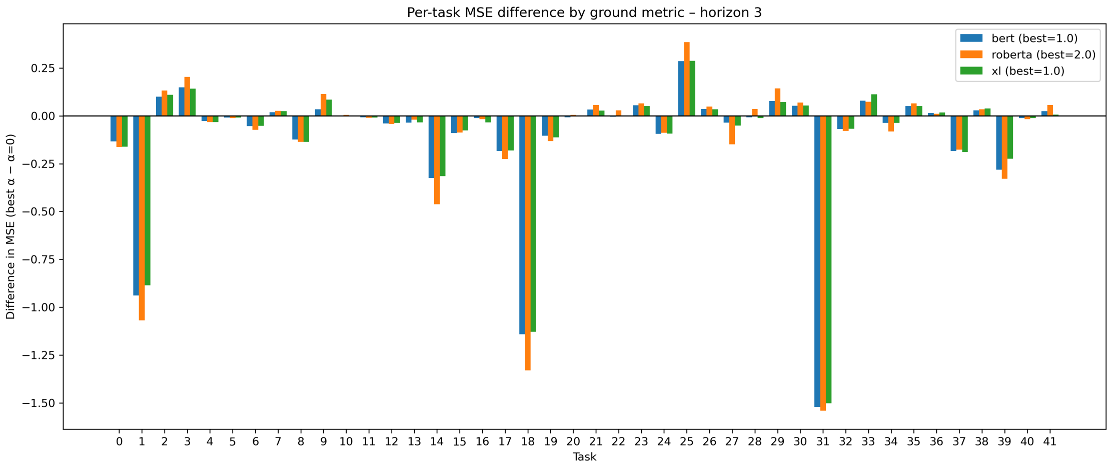

Figure 10

Per-task MSE difference by ground metric for horizon $h = 3$.

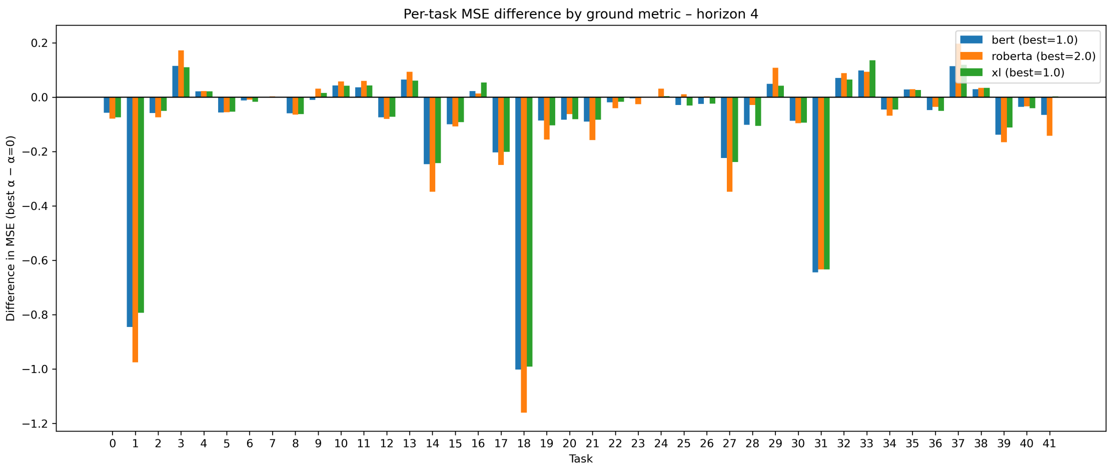

Figure 11

Per-task MSE difference by ground metric for horizon $h = 4$.

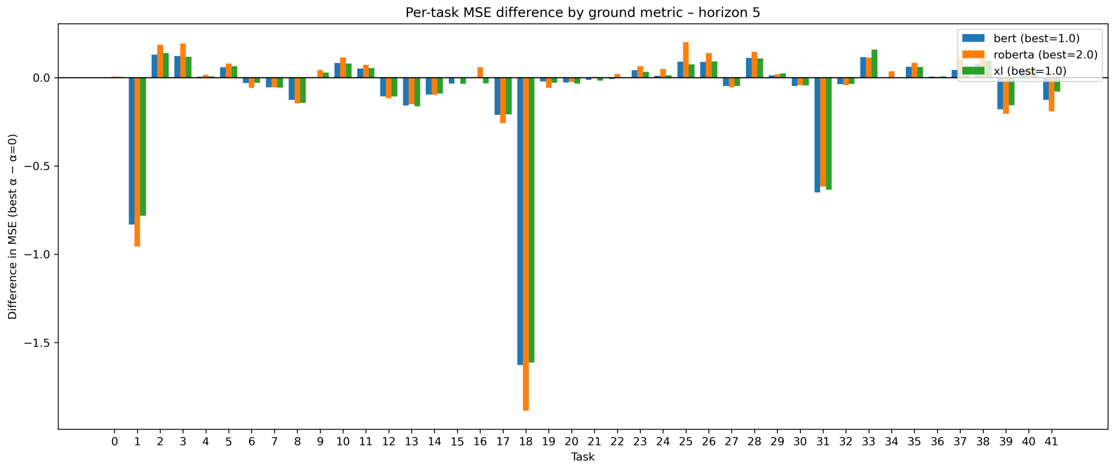

Figure 12

Per-task MSE difference by ground metric for horizon $h = 5$.

Firstly, the majority of tasks exhibit differences clustered near zero across all ground metrics and horizons, reinforcing the earlier finding that the benefits of Wasserstein are modest and consistent. Similar tasks tend to correspond with large benefits throughout horizons, which confirm the outliers seen in the boxplots and show that a subset of participants drive most of the performance gains.

Additionally, the per-task plots show how the horizon length affects the individual trajectories. As the horizon increases, strong improvements become less frequent, which is consistent with error compounding in VAR forecasting. Even at horizon 5, however, a handful of participants still benefit meaningfully from regularization.

Impact of ground metric choice

When comparing ground metrics, the three embedding models exhibit similar distributions for each horizon. Although it is interesting to note that the RoBERTa model chose $\alpha = 2.0$ while the other two chose $\alpha = 1.0$, none of the ground metrics dominate the others in terms of errors. This suggests that the performance gain comes from introducing Wasserstein regularization itself, rather than specific choices of the word embedding models for the distances between variables.

Limitations and Discussion

The present study investigated whether incorporating psychologically and semantically meaningful ground metrics into a Wasserstein regularized multi-task VAR(1) model could improve forecasting accuracy over the standard OLS estimated VAR(1). Across horizons, results generally show modest but consistent gains: median forecast error differences were slightly negative for $h = 1$ through $h = 4$, suggesting that Wasserstein regularization may improve prediction in multivariate EMA time series. Cross-validation consistently selected positive values of α, indicating that the added structure provided by Wasserstein penalty was preferred over the unregularized alternative.

A key finding is the heterogeneity of improvements between individuals. Although aggregate effects are small, per-task analyses show that for a subset of participants, the Wasserstein model substantially reduces prediction error. Consistency across BERT, RoBERTa, and XLNet ground metrics suggest that observed improvements come from the semantically guided regularization, rather than any particular embedding model.

However, there are several limitations of the current analysis. First, the EMA dataset contains missingness that is not random. Participants with missing data may have systematic correlations in their missingness, such as during busy hours or weekends. To mitigate this issue, we excluded participants who had fewer than 87.5% of the 56 total surveys completed. Although this improves data quality, it may contain selection bias by only choosing participants that were more compliant in responding.

Second, for the participants who remained, missing values were imputed using linear interpolation. This preserves the temporal continuity and provides everyone an equal amount of time points, but it could oversmooth the data and not account for abrupt emotional changes at certain days and times.

Third, in choosing the best α value, we used only one held out point for forecasting at each fold. Future work could explore the use of a larger testing set in cross-validation.

There are also limitations inherent to the measurement instruments used in the dataset. The study relies on self reported Likert scale responses, which compress psychological states into numerical categories. These scales are subject to human error, momentary inattention, and may not be perceived equally by participants. This adds noise that any forecasting model must accommodate, which may explain modest improvements.

Despite these limitations, the findings suggest that incorporating semantic structure through a Wasserstein penalty offers a promising direction for improving psychological time series modeling. Future work could explore richer ground metrics, increased lags, and a variety of missing data handling.

Overall, the study demonstrates that regularization based on optimal transport can incrementally improve VAR forecasts while highlighting the heterogeneous nature of emotions across individuals. The results offer conceptual and methodological value for future research, and can serve as a step towards integrating semantic structure into multivariate time series modeling.

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